

SP250 Homework 2

1. Let $n > 1$ be a number less than 10,000 that has no divisors greater than 1 and less than 100. Prove that n is a prime number.

Solution:

2. What is the sum of the first 100 numbers that are not divisible by 6?

Solution:

3. Is it possible to connect 11 towns with roads in such a way that every town is connected to exactly 3 others?

Solution:

4. a) Find all prime numbers p such that $p^2 + 3$ is also a prime.
b) Find all prime numbers p such that $p^2 + 2$ is also a prime.

Solution:

5. Joey has a gallon of wine and a gallon of water. He pours a pint of wine into water and then a pint of the mixture back into the container with wine. Does he now have more water in the wine container or more wine in the water container?

Solution:

6. Two players play a game. The first player puts a chess rook anywhere on the board. After that, players alternate by moving the rook according to rules of chess, starting with the second player. It is forbidden to move the rook into a square that has already been visited, including the square where the rook was originally placed. The player who makes the final move wins. If both players play optimally, does one of the players have a winning strategy, and if so – which one?

Solution:
